14.7 Answers

(1) a) $f(1,1) = f(-1,-1) = -1$ is a local min.

$(0,0,1)$ is a saddle point.

b) $f(2,0) = e^4$ local max

c) $f(0,0) = 0$ local min, $f\left(0,-\frac{5}{3}\right) = \frac{125}{27}$ local max

$(\pm 2,-1,3)$ saddle pts.

d) For every integer k, $(k\pi,0,0)$ is a saddle point,

e) $f(1,1) = f(-1,-1) = 4$ local min

(2) a) $f(0,2) = -9$ abs min

$f(3,0) = 10$ abs max

b) $f(0,0) = 6$ abs min

$f(1,1) = 9$ abs max

c) $f(1,1) = 1$ abs min

$f(3,0) = 84$ abs max

d) $\boxed{f(0,y) = f(x,0) = 0 \text{ abs min}}$ where $0 \le x,y \le 2$.

$\boxed{f\left(\sqrt{\frac{8}{3}}, \frac{2}{\sqrt{3}}\right) = \frac{16}{3\sqrt{3}} \text{ abs max}}$

(3) $\left(\frac{10}{3}, \frac{7}{3}, \frac{5}{3}\right)$

(5) $20\sqrt[3]{7} \times 20\sqrt[3]{7} \times 10\sqrt[3]{7}$ cm

(4) $(0, \pm\sqrt{2}, 0)$